

Report: Battery Classification For Proper Disposal

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Dataset Overview

The dataset used is a custom dataset containing a number of battery types, these types are Alkaline, Li-ion, Lithium, Ni-CD, Ni-MH. This dataset is made from scrapping a multitude of sources which I have included in the github repository. The dataset was preprocessed by resizing all images to 224x224, and splitting the dataset into training (80%), validation (20%), and a separate test set for evaluation.

Model Choice

For this project, a pre-trained ResNet-18 model was selected and fine-tuned using PyTorch. The pre-trained model was initialized with ImageNet weights to leverage transfer learning, which significantly improves accuracy while reducing the need for large training datasets.

The architecture consists of multiple convolutional layers, with only the last two blocks (layer 3 and layer 4) being unfrozen for fine-tuning. A custom classifier replaces the default fully connected layer to adapt the model for battery classification.

The classifier includes:

- A dropout layer (50%) to prevent overfitting.
- A fully connected layer reducing features to 512 dimensions.
- GELU activation for better non-linearity.
- Batch normalization to stabilize training.
- A final linear layer that maps features to the five battery types.

This approach balances efficiency and performance by utilizing a robust, pre-trained feature extractor while allowing domain-specific learning through fine-tuning.

Model Operability

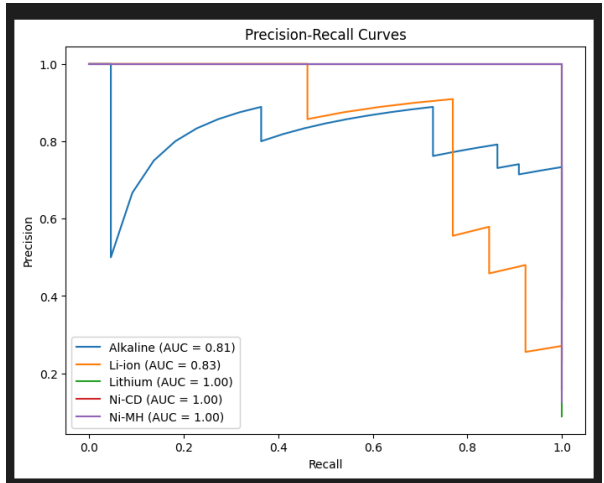
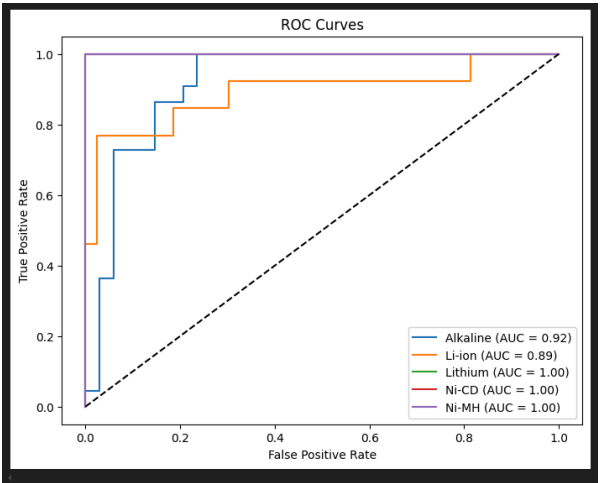
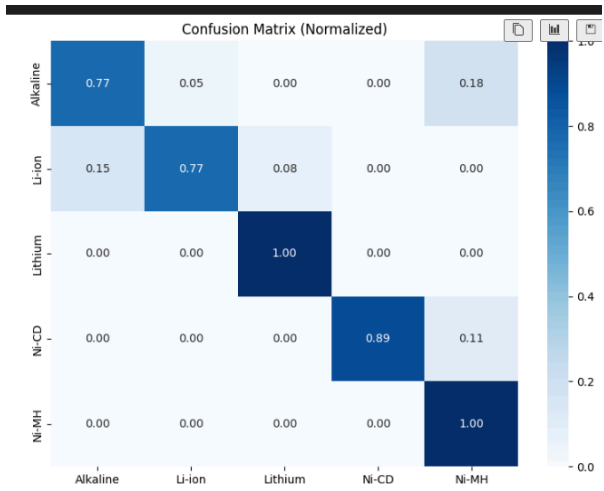
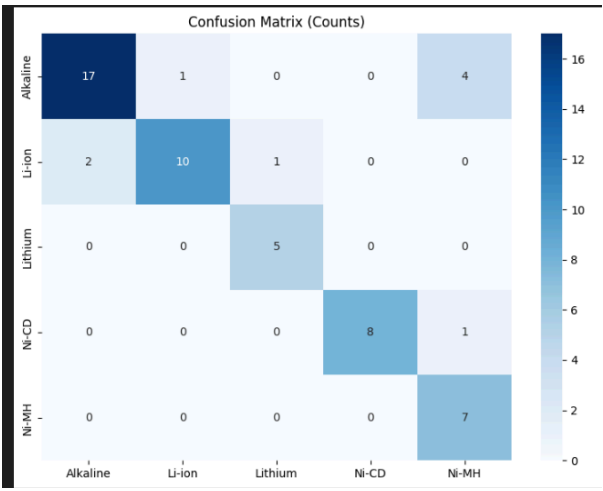
To ensure accessibility and usability, the trained model has been containerized using Docker. This allows the classification model to be deployed across various environments without dependency conflicts.

Key Performance Metrics

The model achieved an accuracy of 84% on the test dataset. The following metrics were used to evaluate its performance:

- Precision: Measures the percentage of correct positive predictions.
- Recall: Evaluates the model's ability to correctly identify battery types.
- F1-Score: Balances precision and recall for overall effectiveness.
- Confusion Matrix:
- ROC Curve: Shows the trade-off between true positive rate and false positive rate
- AUC Curve: Measures the overall ability of the model to distinguish between classes.

Performance Metrics



Classification Report:				
	precision	recall	f1-score	support
Alkaline	0.895	0.773	0.829	22
Li-ion	0.909	0.769	0.833	13
Lithium	0.833	1.000	0.909	5
Ni-CD	1.000	0.889	0.941	9
Ni-MH	0.583	1.000	0.737	7
accuracy			0.839	56
macro avg	0.844	0.886	0.850	56
weighted avg	0.871	0.839	0.844	56

Insights & Observations

The model's performance demonstrates a strong capability in distinguishing between different battery types, with an overall accuracy of 84%. However, misclassifications still occur, particularly due to the inherent difficulty of classifying battery types solely based on visual appearance, especially when there is no text to indicate the type.

A confusion matrix heatmap was utilized to visually analyze classification errors, further confirming that the model performs better on batteries with more visually distinguishable features. Smaller sample sizes for certain battery types most likely contributed to lower recall, meaning it might be beneficial to increase the dataset.

Finally, chaining two models together may enhance results, as some battery types are more difficult to differentiate than others.

Conclusion

The model with PyTorch successfully classifies batteries with over 84% accuracy.